

Question	Answer	Marks
5(a)	any 2 points from: • (angular) displacement • velocity • momentum • (centripetal) acceleration • (resultant) force	B2
5(b)(i)	$Bqv = mv^2 / r$	M1
	$v = 2\pi r / T$	M1
	completion of algebra leading to $B = 2\pi m / qT$	A1
5(b)(ii)	$B = (2\pi \times 4 \times 1.66 \times 10^{-27}) / (2 \times 1.60 \times 10^{-19} \times 2.5 \times 10^{-6})$	C1
	$= 0.052 \text{ T}$	A1
5(b)(iii)	either the same because T is independent of r or the same because B, q and m are unchanged or the same because both radius and speed have doubled	B1
5(b)(iv)	$qE = Bqv$	C1
	$E = Bv = 0.052 \times 1.1 \times 10^6$ $= 5.7 \times 10^4 \text{ N C}^{-1}$	A1

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